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Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature

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With special thanks to Behrad Khorramnazari (The Synthesis Company of California) for his work on this project.



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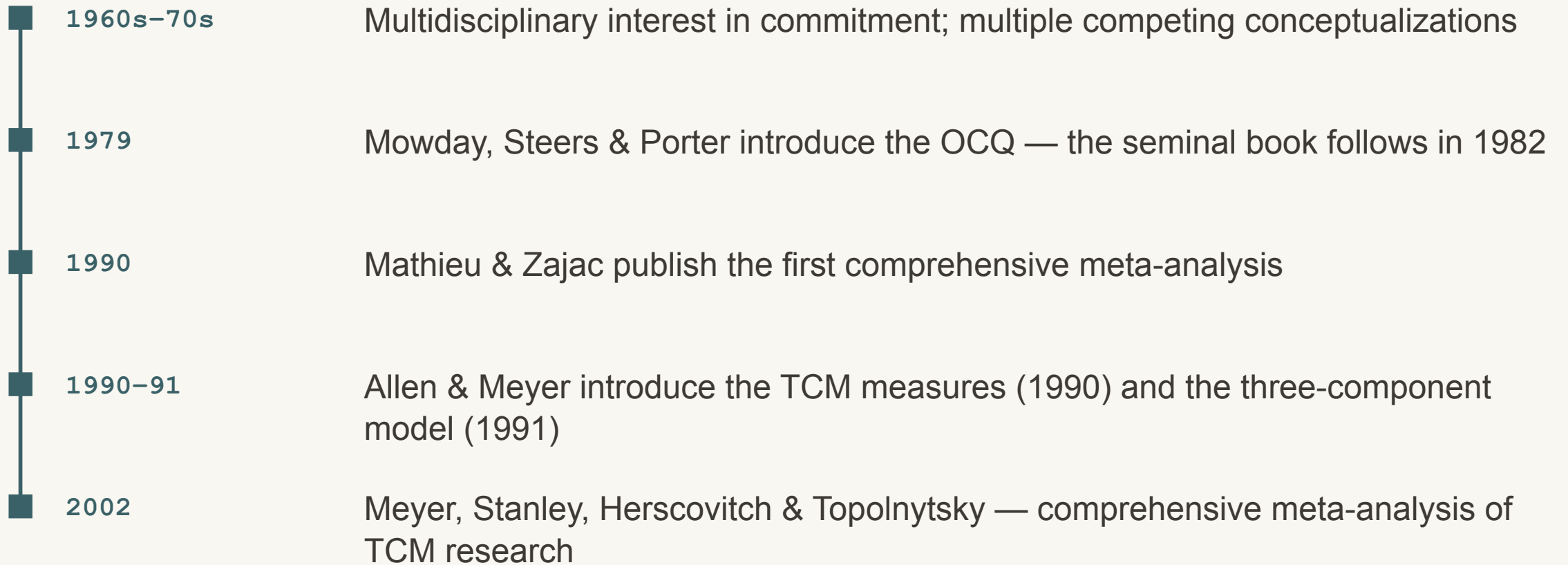
Agenda

Where we've been, where we are, and where this is going

- Brief History of Commitment Theory & Research
- Challenges to conducting comprehensive reviews
- Toward an AI-Assisted Meta-Analysis
 - Where we are
 - Where we are going
 - What is possible

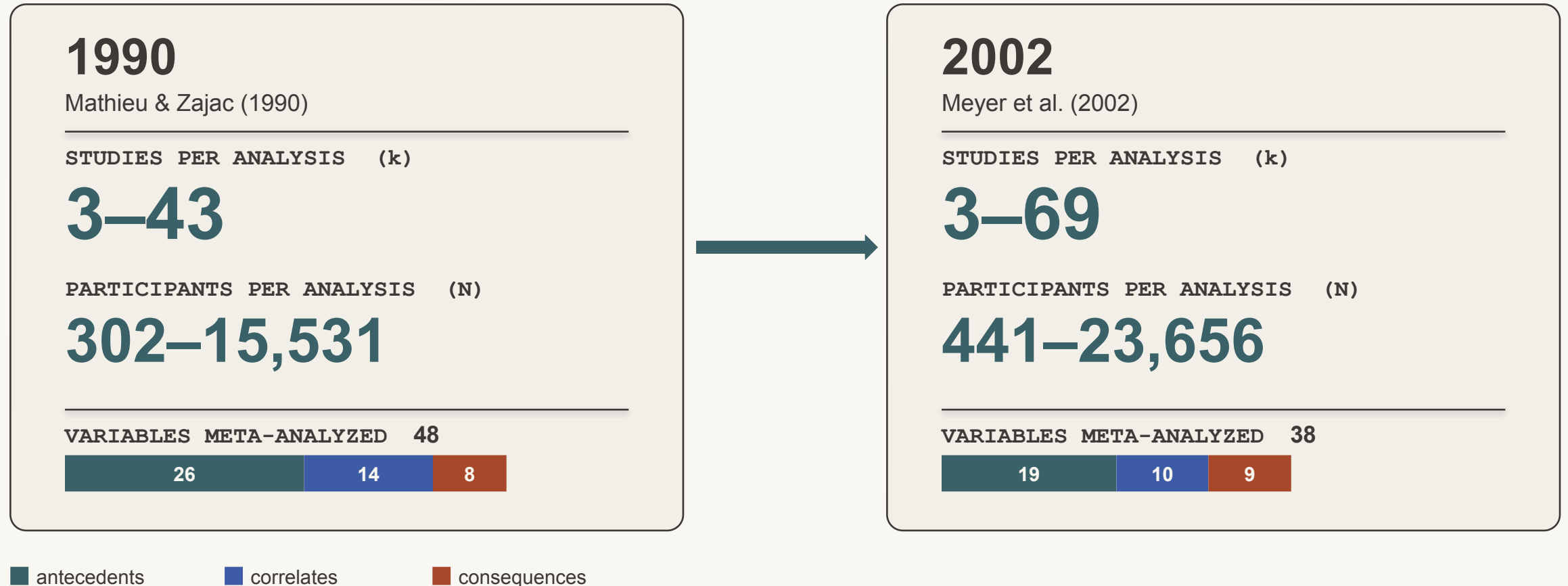
Organizational commitment: the early years

A brief history — some milestones



Two comprehensive meta-analyses

Studies (k) and participants (N) per analysis — Mathieu & Zajac (1990) and Meyer et al. (2002)

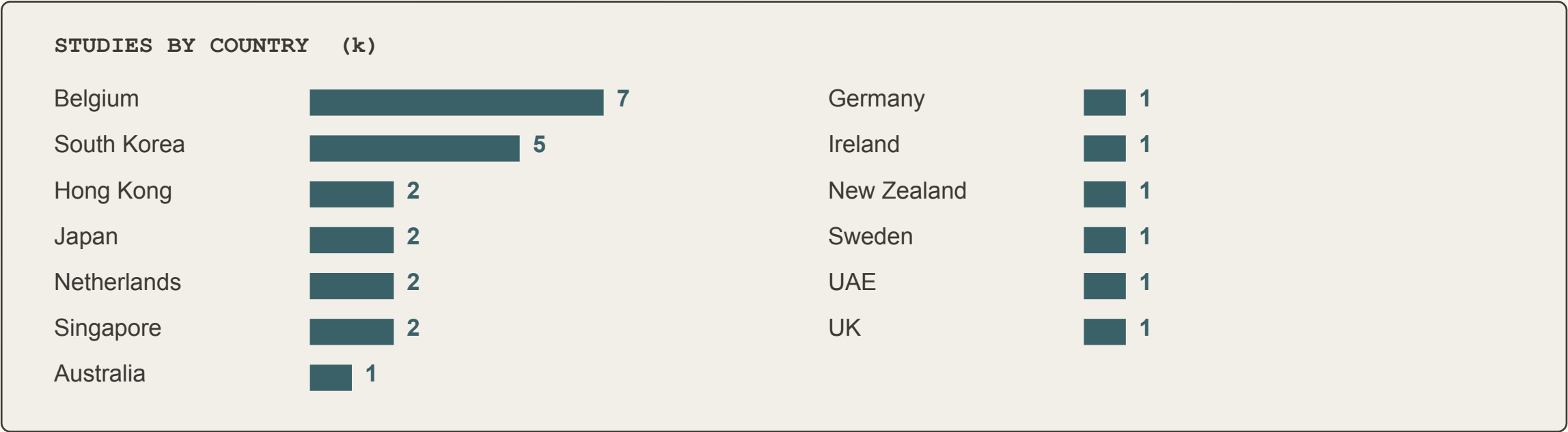


A modest cross-national base

Meyer et al. (2002) — studies reporting country of origin

■ GEOGRAPHIC REACH OF THE EVIDENCE

13 countries · 27 studies



Studies reporting country of origin (n = 27); tabulation from the ICAP 2014 presentation.

Commitment across cultures

Meyer et al. (2012) — studies (k) and participants (N) per commitment component

■ NORMATIVE DATA ASSEMBLED FROM

54 countries · **9** geographic regions

AC

Affective commitment

STUDIES (k)

966

PARTICIPANTS (N)

433,129

CC

Continuance commitment

STUDIES (k)

428

PARTICIPANTS (N)

199,831

NC

Normative commitment

STUDIES (k)

336

PARTICIPANTS (N)

133,277

Challenges to conducting comprehensive reviews

Why a truly comprehensive commitment review has become impractical by hand.

1

Studies keep growing

Commitment research piles up faster every decade.

2

Reviews turn narrow

Meta-analyses cover fewer variables to stay feasible.

3

Even narrow is too much

Now even targeted reviews outstrip human effort.

4

AI shares the labour

AI screens and codes at scale, clearing the bottleneck.

Literature search method

Two complementary search strategies in every database (except ERIC)

- **Keyword & subject-term search** — truncated organizational terms (org*, work*, employ*, job*, firm*, institution*, affect*, norm*, continu*) within three words of commit*, plus each database's subject headings; syntax adapted to each platform.
- **Cited-reference search** — every record citing any of 14 foundational commitment works (e.g., Porter et al., 1974; Mowday et al., 1979; Allen & Meyer, 1990; Meyer & Allen, 1991; Meyer et al., 2002; Klein et al., 2014).
- **Scholarly sources only** — academic journals, books, dissertations & theses, conference papers, reports, and working papers.

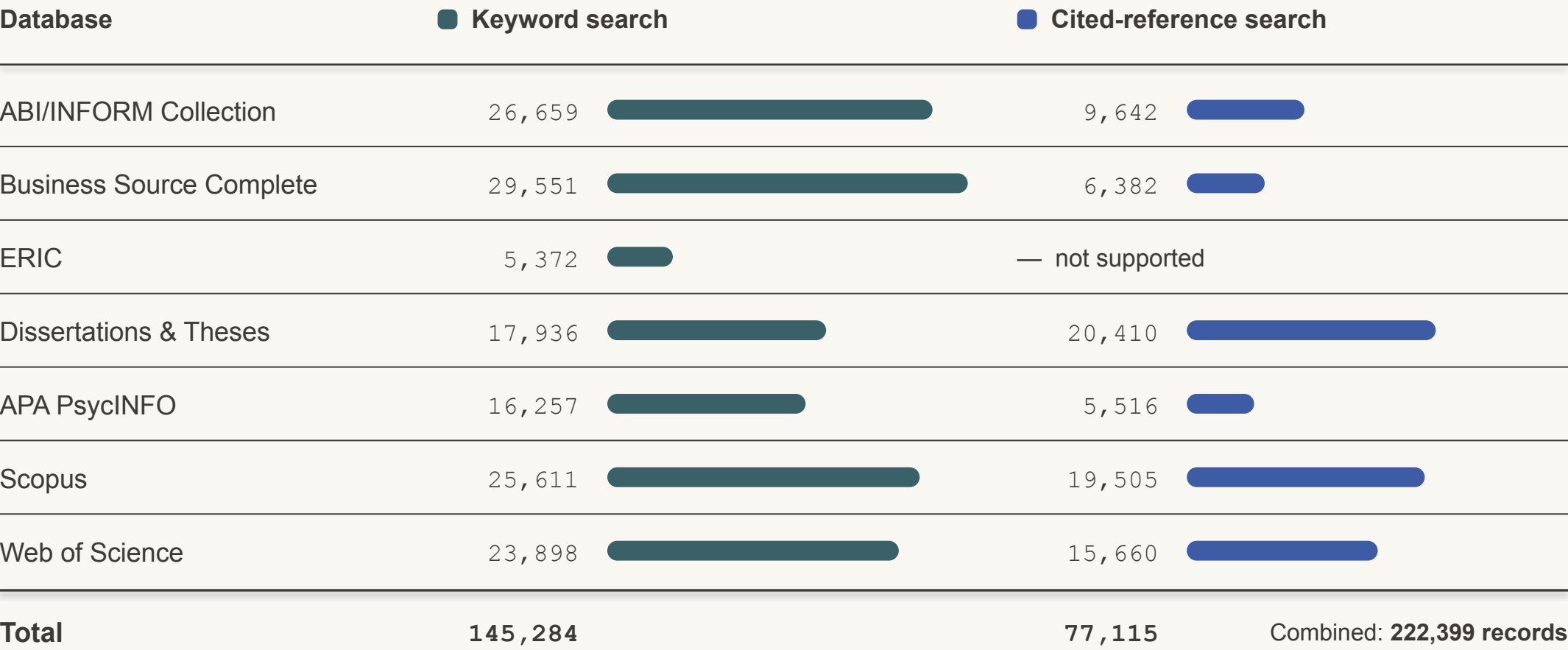
Seven databases

Searched December 20, 2025 – January 7, 2026

ABI/INFORM Collection	ProQuest
Business Source Complete	EBSCO
ERIC	Ovid · keyword only
Dissertations & Theses	ProQuest
APA PsycINFO	Ovid
Scopus	Elsevier
Web of Science	Clarivate

Records identified, by database and search strategy

Records retrieved per database, before consolidation and de-duplication · bars share one scale



ERIC (Ovid) does not support cited-reference searching.

From 222,399 records to 22,484 unique abstracts

All records retrieved — 13 searches across 7 databases

222,399



▼ consolidated and de-duplicated



22,484

unique abstracts

Selected by AI screening for relevance

10.1% of the initial yield — the abstracts behind every analysis that follows

How accurate is AI screening? — from prior research

Before any coding, every record has to be judged relevant or not — the first filter, deciding which studies even enter the review. In earlier published validation, the AI was compared against the expert human teams who screened by hand.

■ Screening — deciding which studies belong

97.2%

of relevant studies — vs 91.6% for humans

5 meta-analyses · 30,150 abstracts screened

■ What the AI is being compared to

The human baseline was thorough — every record judged by hand:

2×

two reviewers + a 3rd

→

1 pass

the AI

≈ 98–99% less time — days or months of work, done in minutes to an hour.

[Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. \(2026\). *Improving Evidence Synthesis with Artificial Intelligence* \[Preprint\].](#)

WHAT THE CODED ABSTRACTS REVEAL

Abstract Analysis

How the literature has grown — and what it studies.



How quickly has the literature on organizational commitment grown? Here's the annual count of articles in our collection.

Articles Published Per Year

Annual count of unique abstracts, 1970–2025 (n = 21,753 of 22,484)



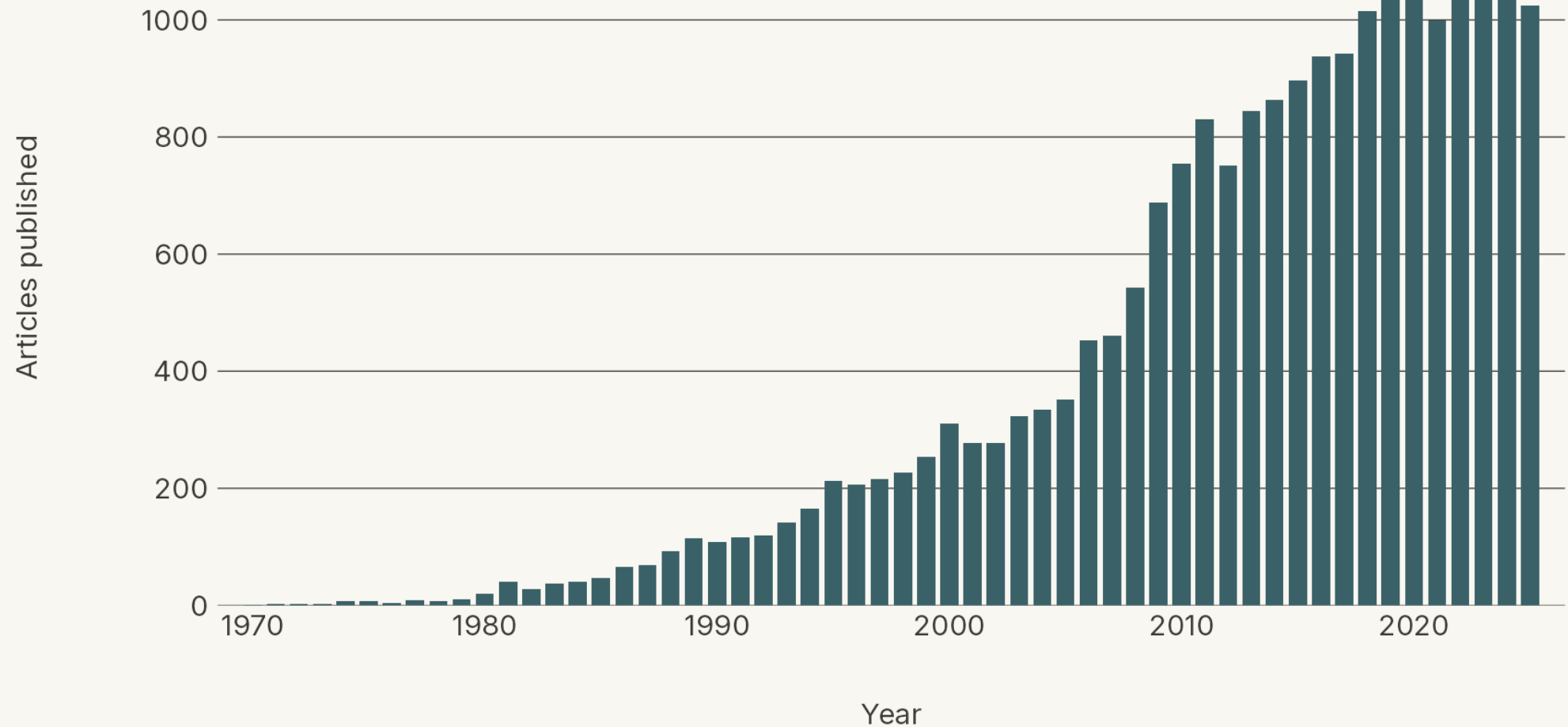
Articles Published Per Year

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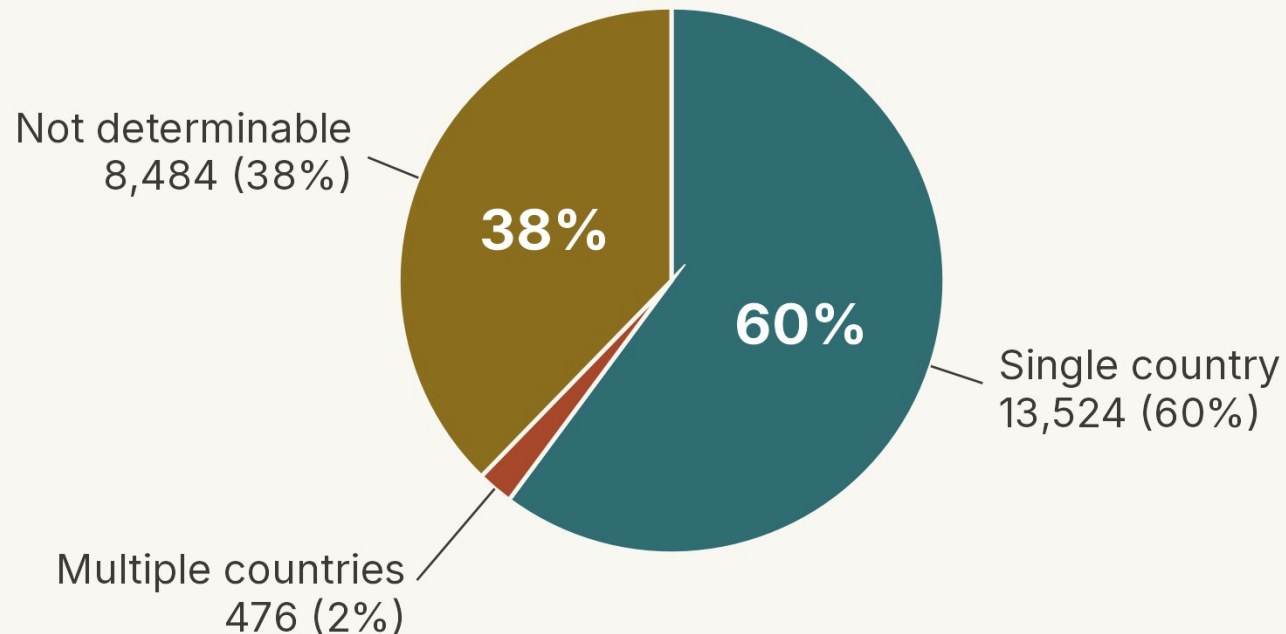




And mapped onto the globe — the geographic spread of the literature over five decades.

How many of the articles can we place geographically?

Country-coding status across all 22,484 unique abstracts



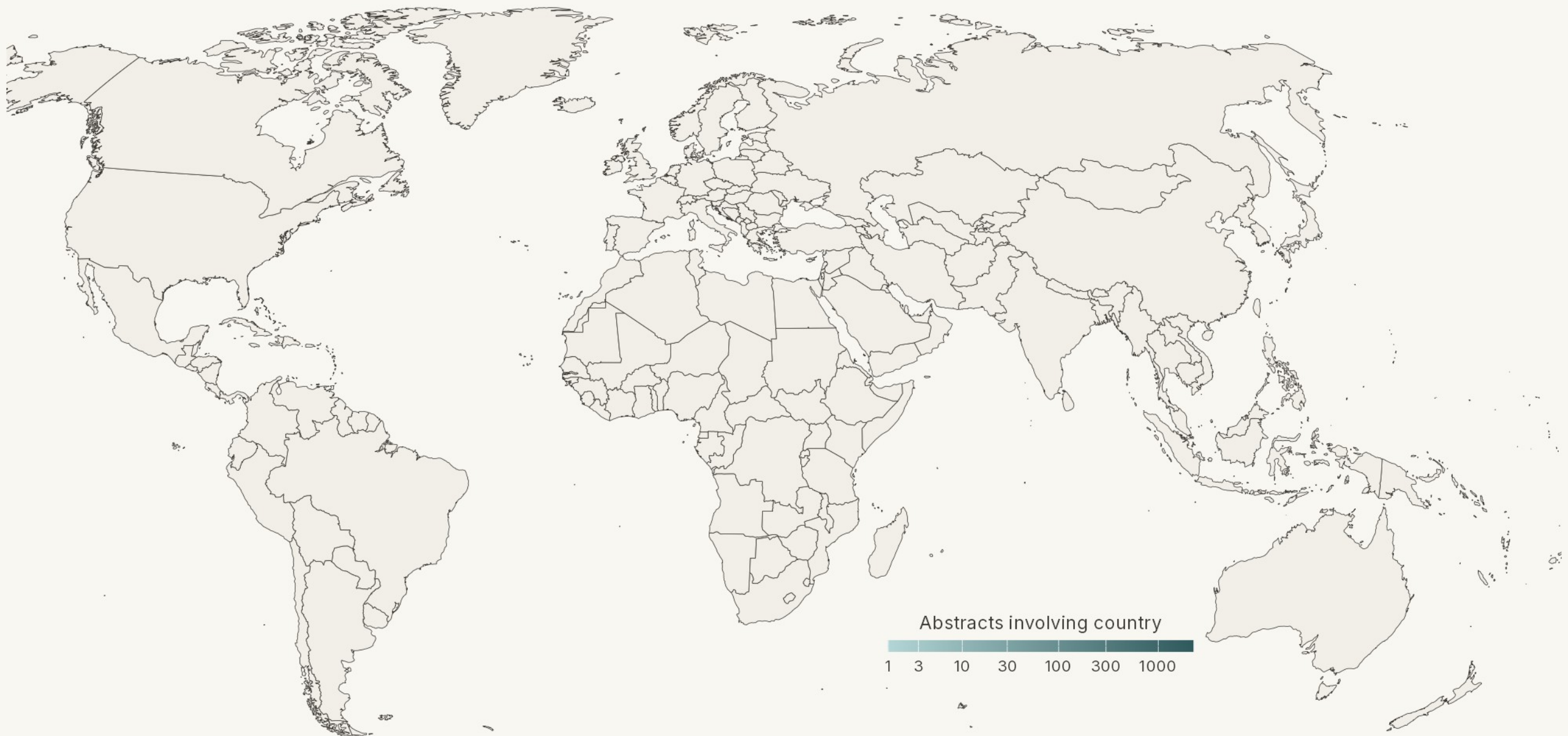
How country of origin was determined

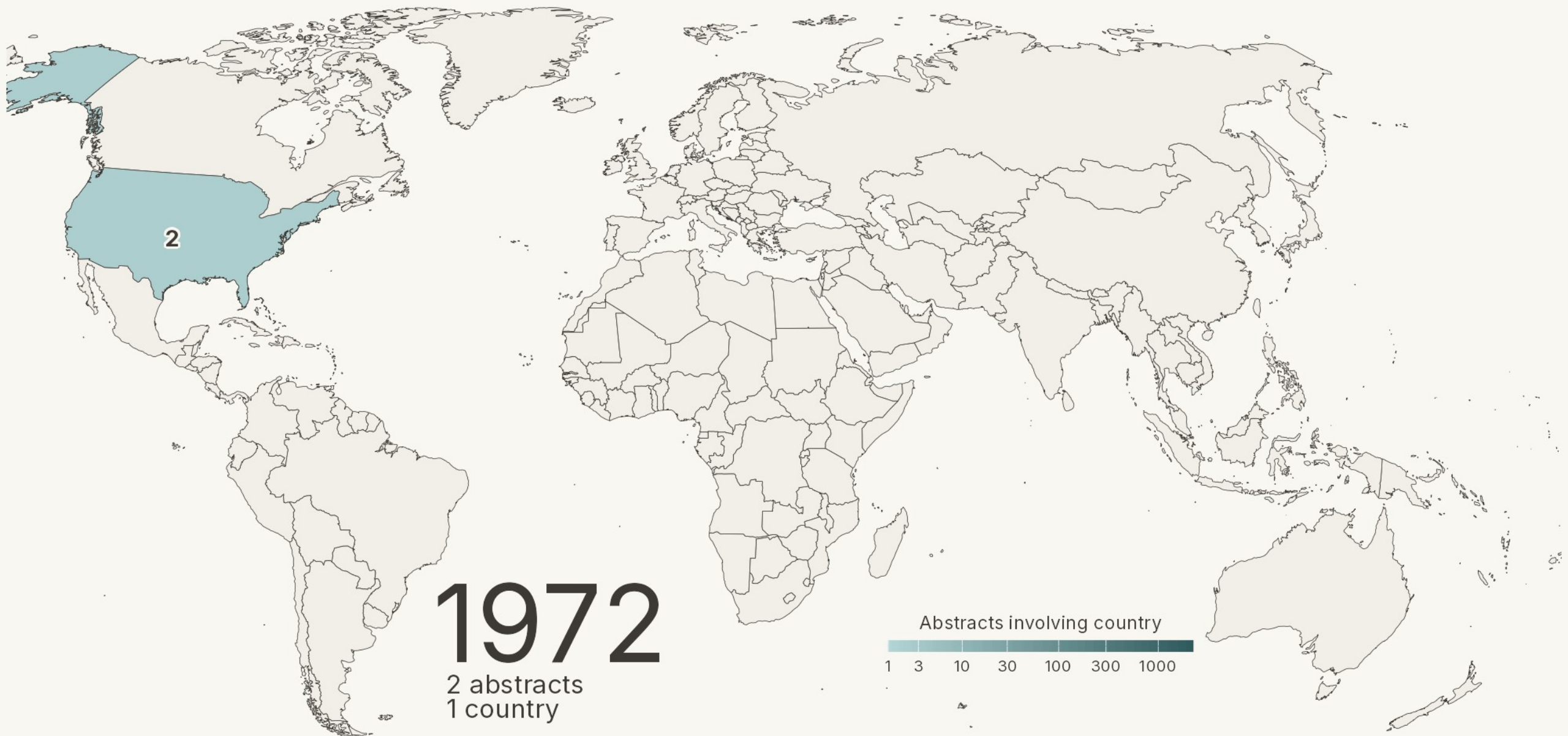
Country coding criteria (each YES / NO):

- Is a country named directly in the abstract?
- Does a stated city or region map to a country?
- Does a named company or organization map to a country?
- Do all authors' affiliations point to the same country?

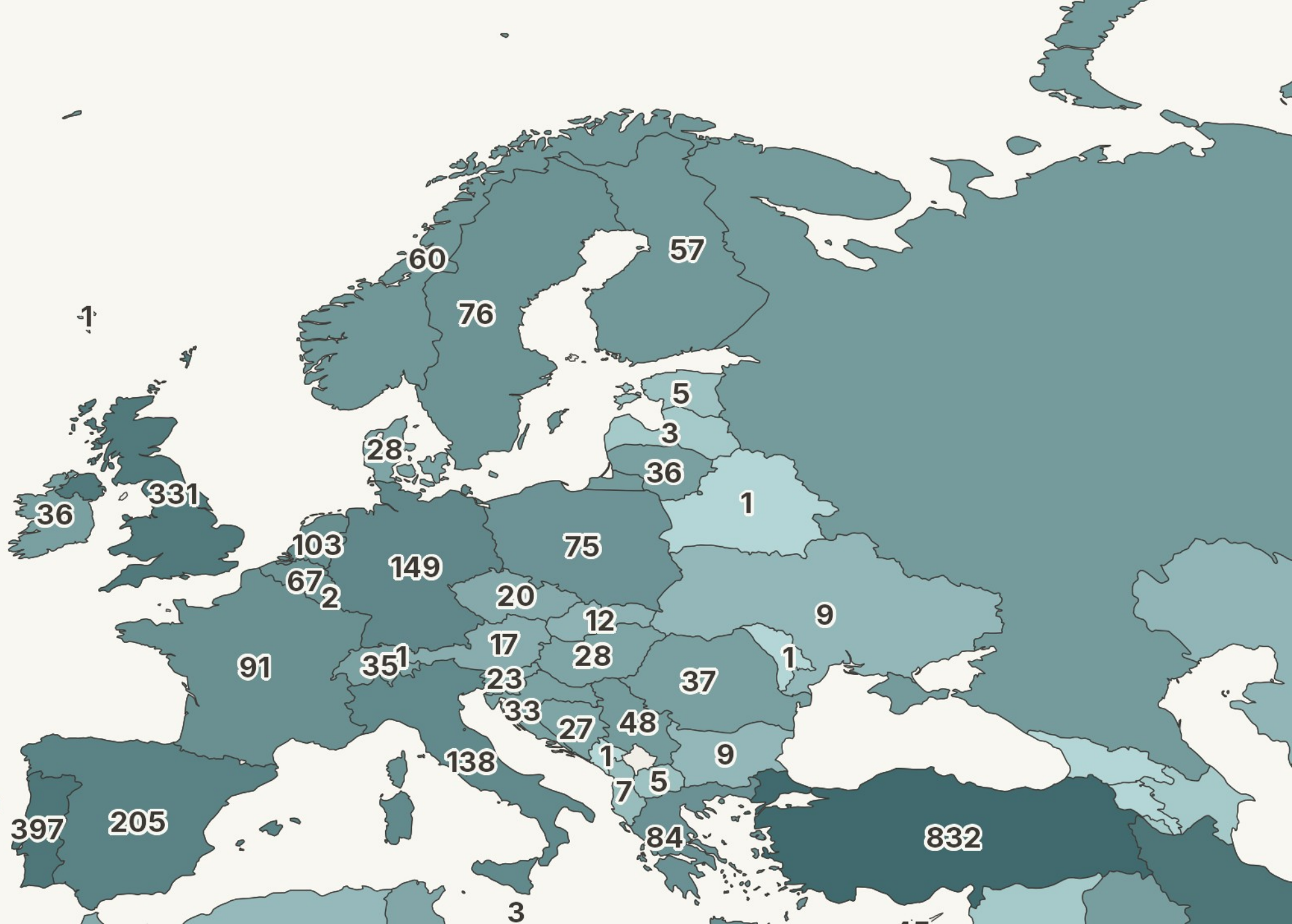
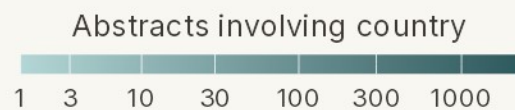
Recovering undetermined cases:

- For undetermined cases, author affiliations were retrieved from OpenAlex via DOI lookup and passed to the model, resolving 14 previously undetermined items.



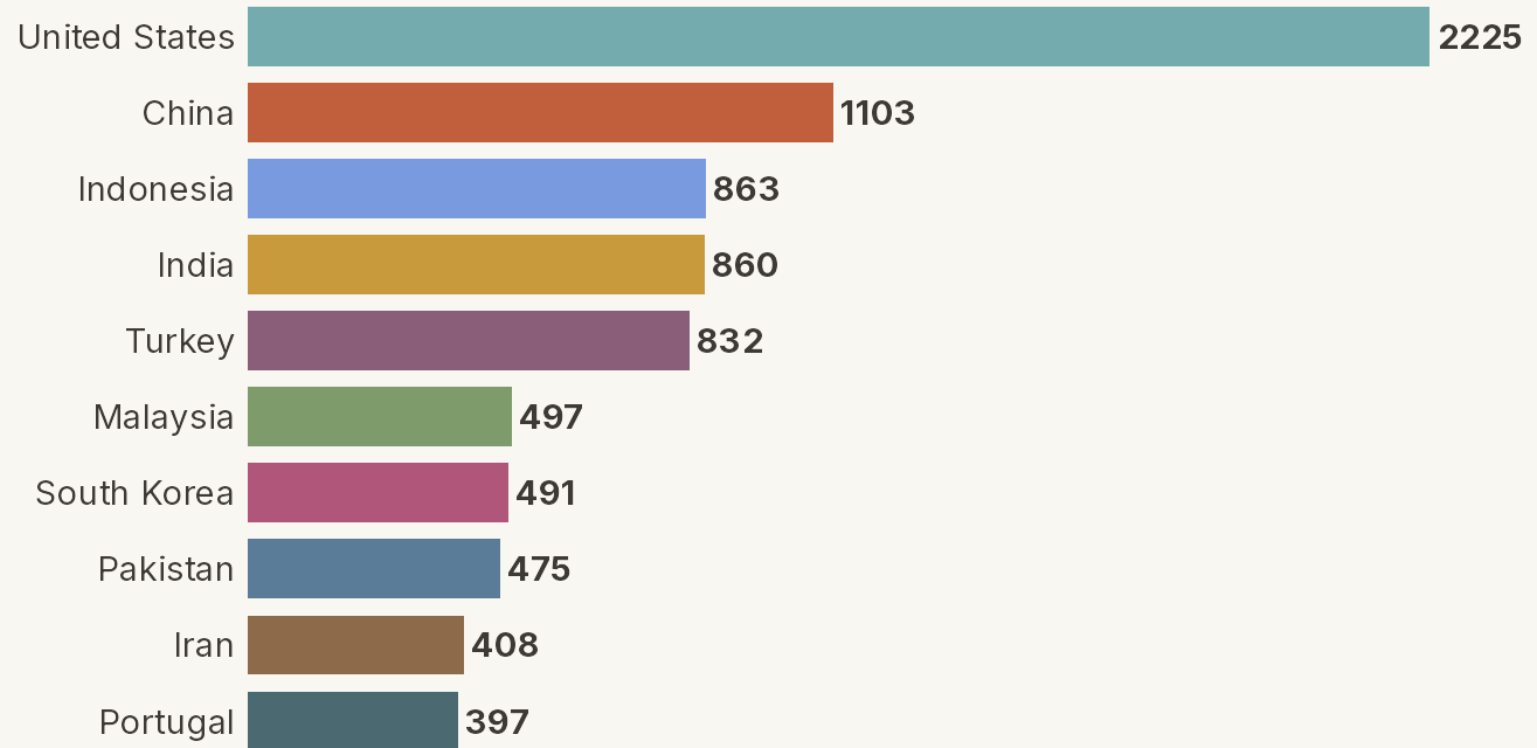


2025



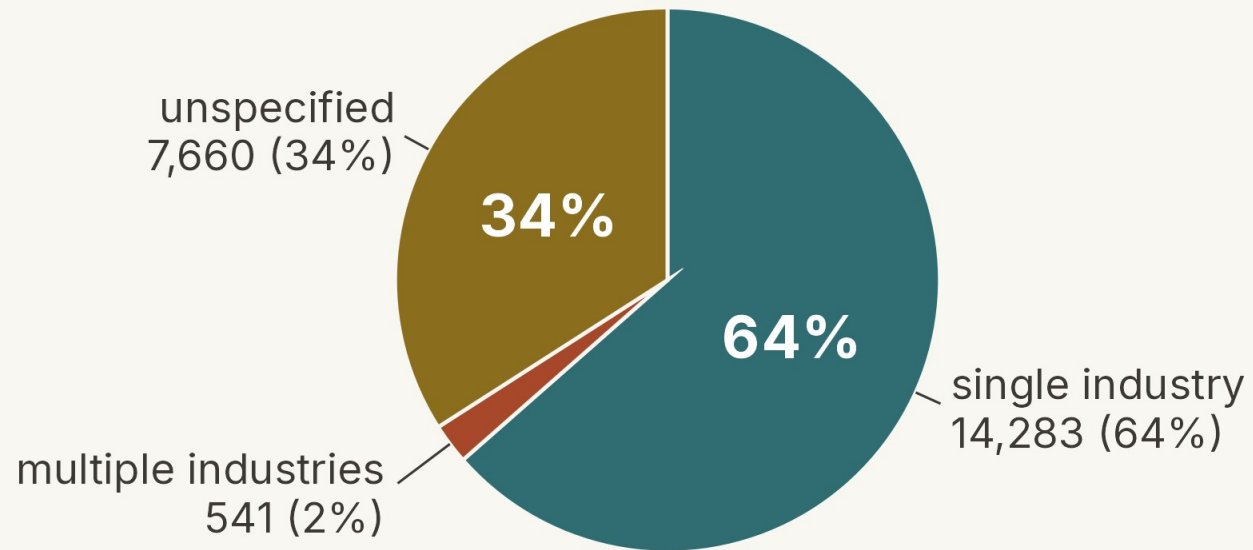
Top 10 Countries by Cumulative Samples

2025



Industry — coverage in three tiers

Full coded sample, N = 22,484



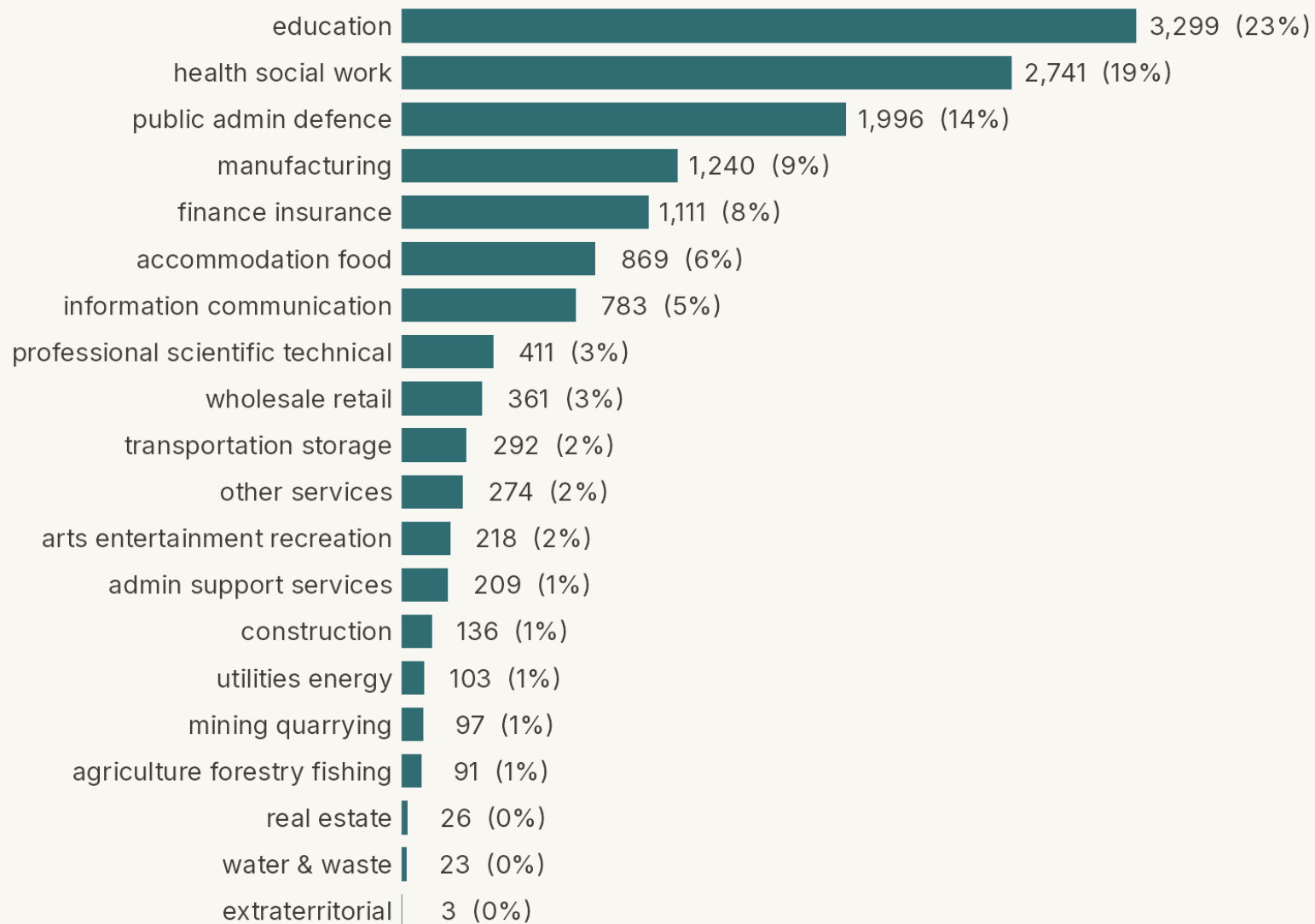
Industry taxonomy — ISIC Rev. 4

The UN's standard industry taxonomy — 21 letter-coded sections, A (agriculture, forestry & fishing) through U (extraterritorial organizations).

[Taxonomy Details Website](#)

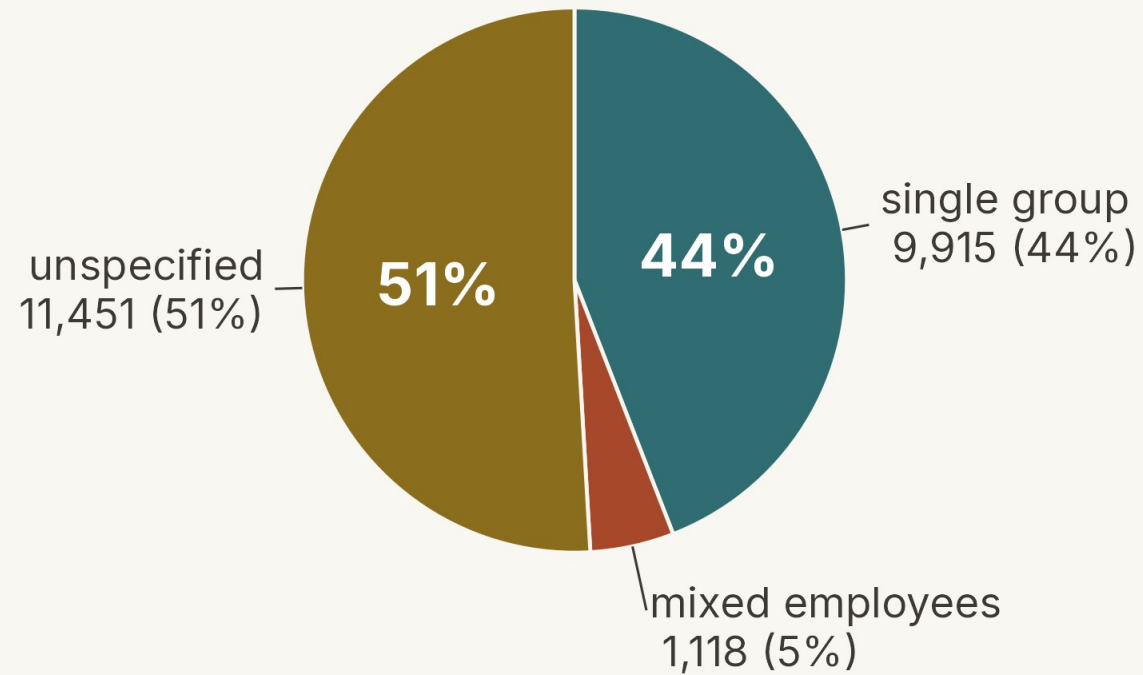
Industry — distribution among placed studies

Single-industry studies, N = 14,283



Occupation — coverage in three tiers

Full coded sample, N = 22,484



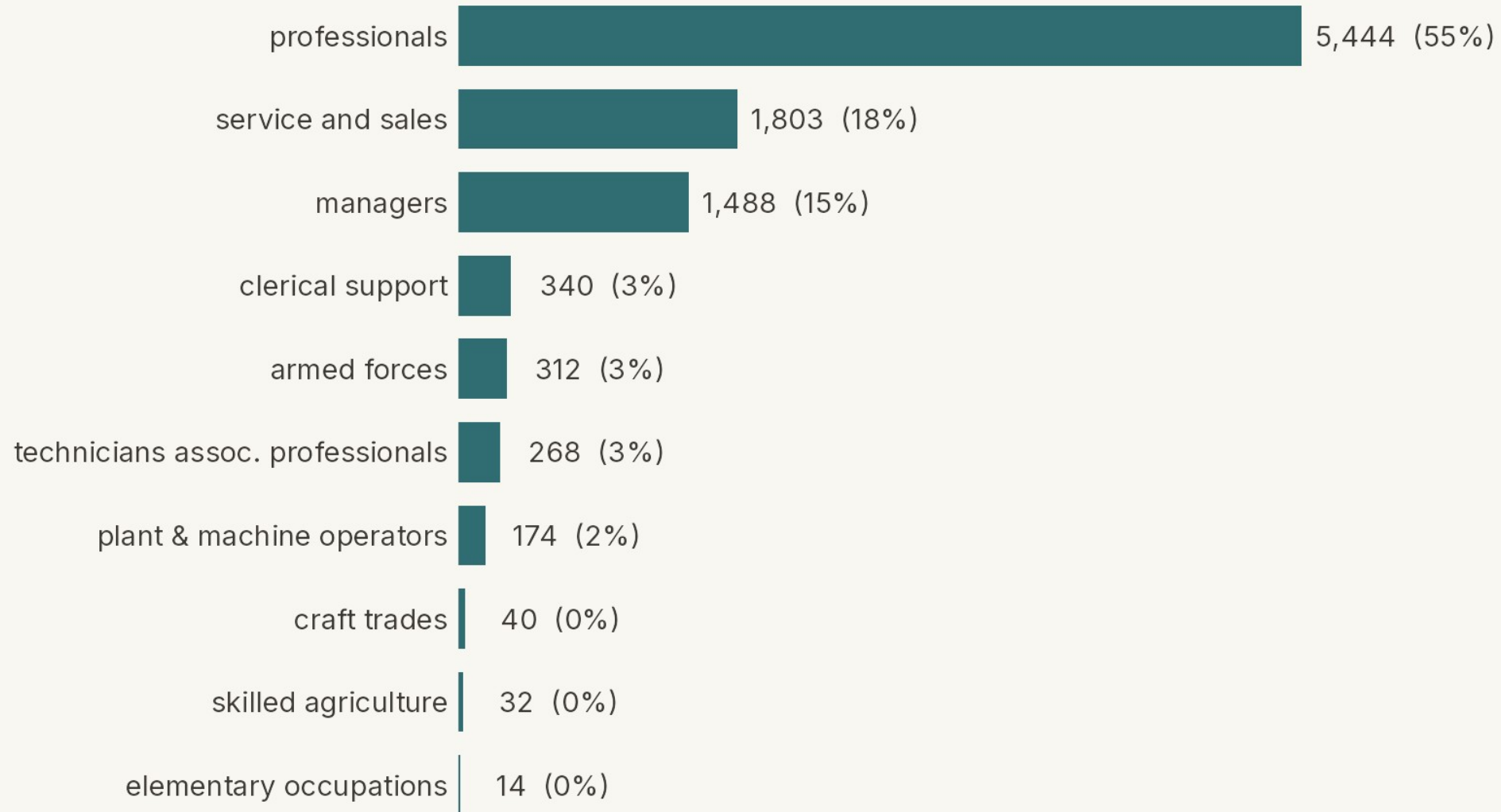
Occupation taxonomy — ISCO-08

The ILO's standard occupation taxonomy (the counterpart to ISIC) — 10 major groups, 0 (armed forces) through 9 (elementary occupations).

[Taxonomy Details Website](#)

Occupation — distribution among single-group studies

Single-group studies, N = 9,915



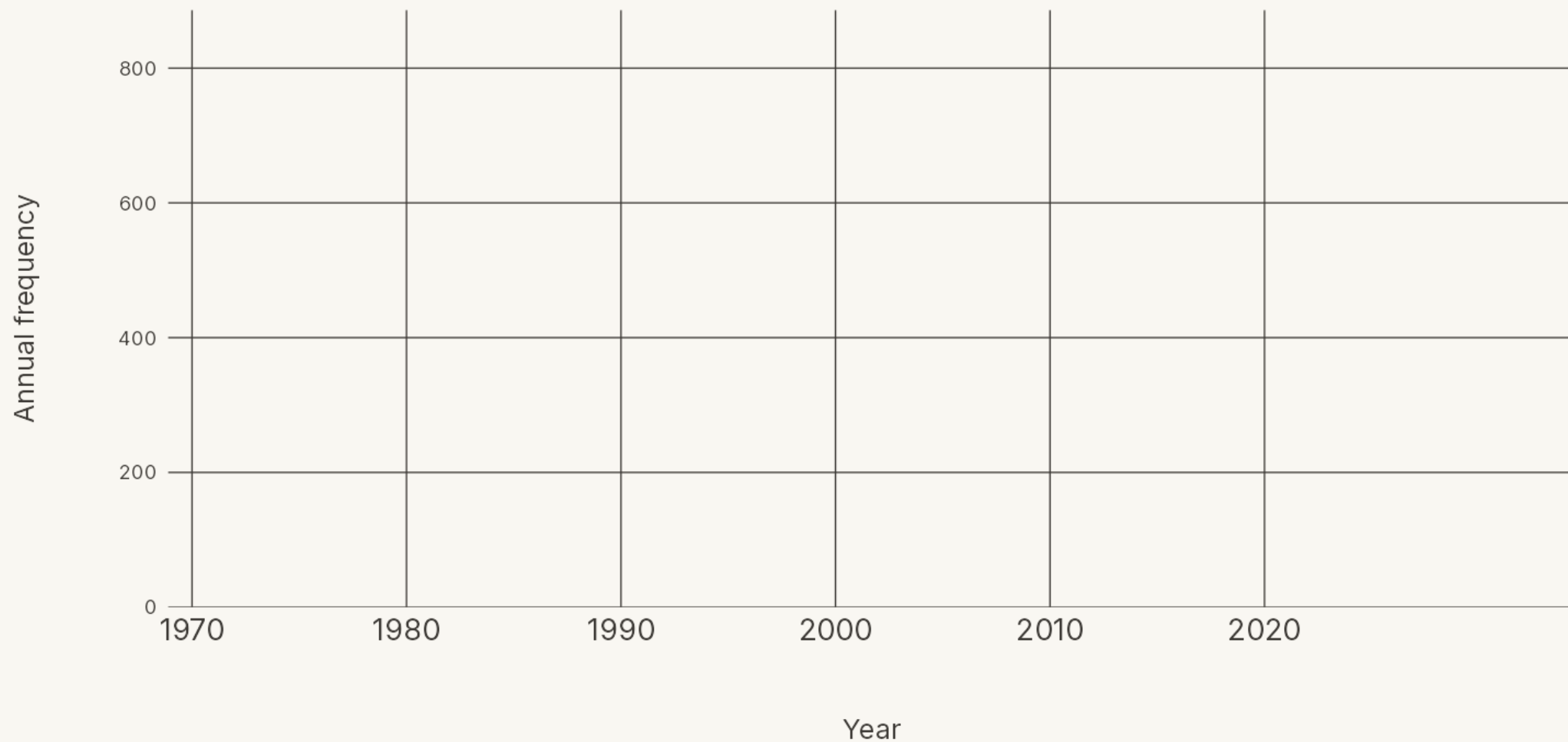
Finally, how is “organizational commitment” actually used in the literature? It plays four primary roles — as a predictor, a criterion, a correlate, and a mediator — and the balance has shifted over time.

How the four roles were assigned

We trained the AI to build a path model from each abstract's text. A variable's position in that path model determined which of the four roles it plays.

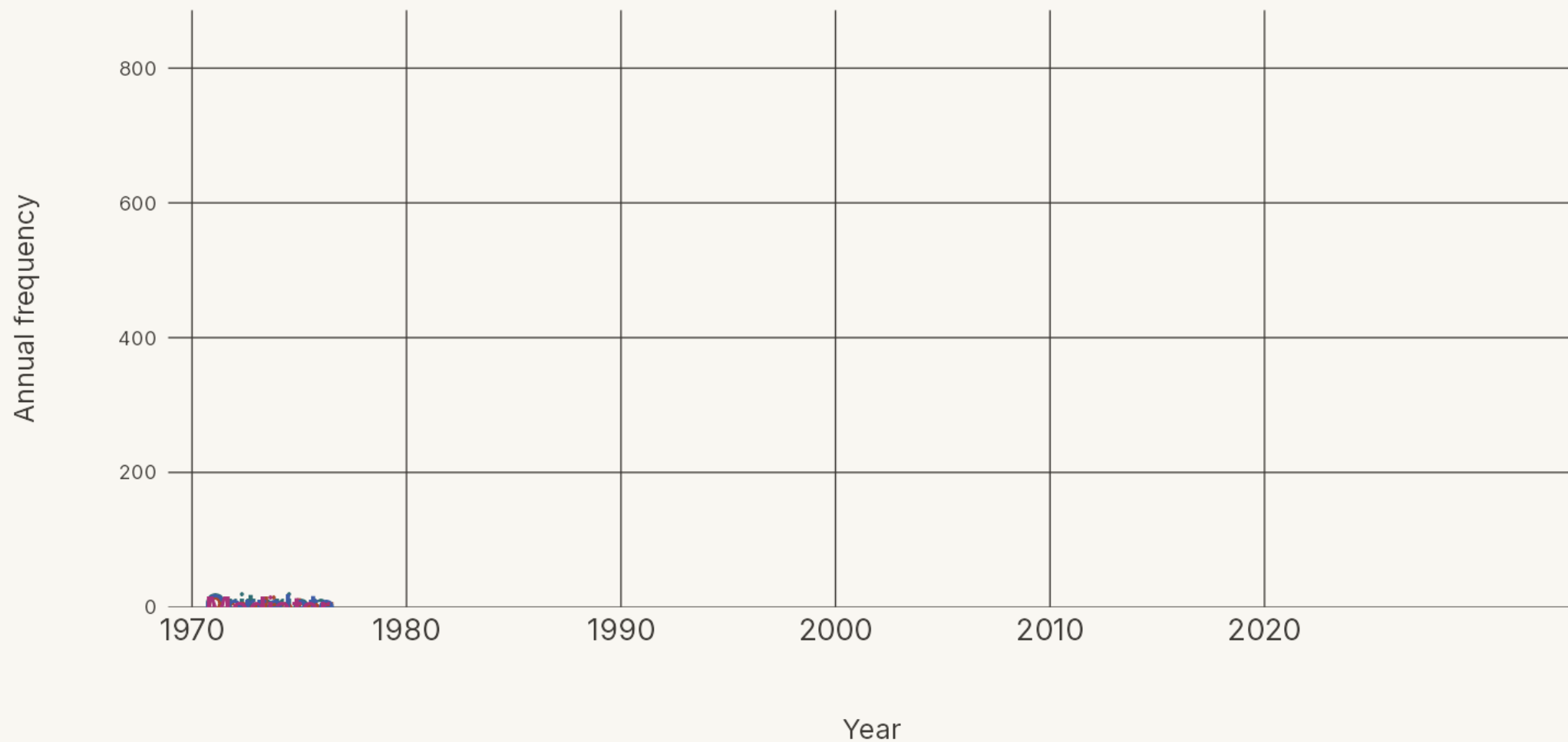
How 'organizational commitment' is used

Annual count of abstracts using commitment in each role



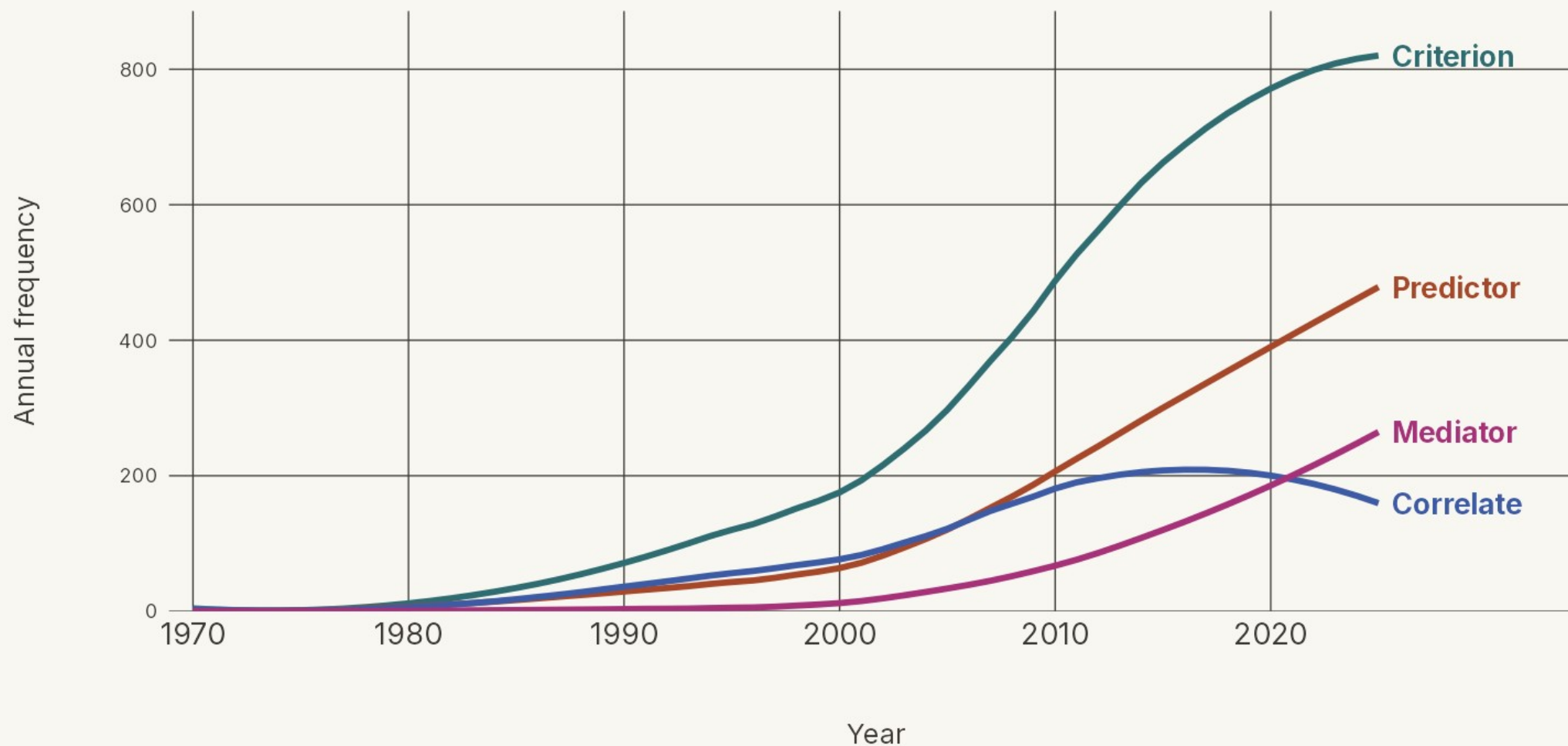
How 'organizational commitment' is used

Annual count of abstracts using commitment in each role



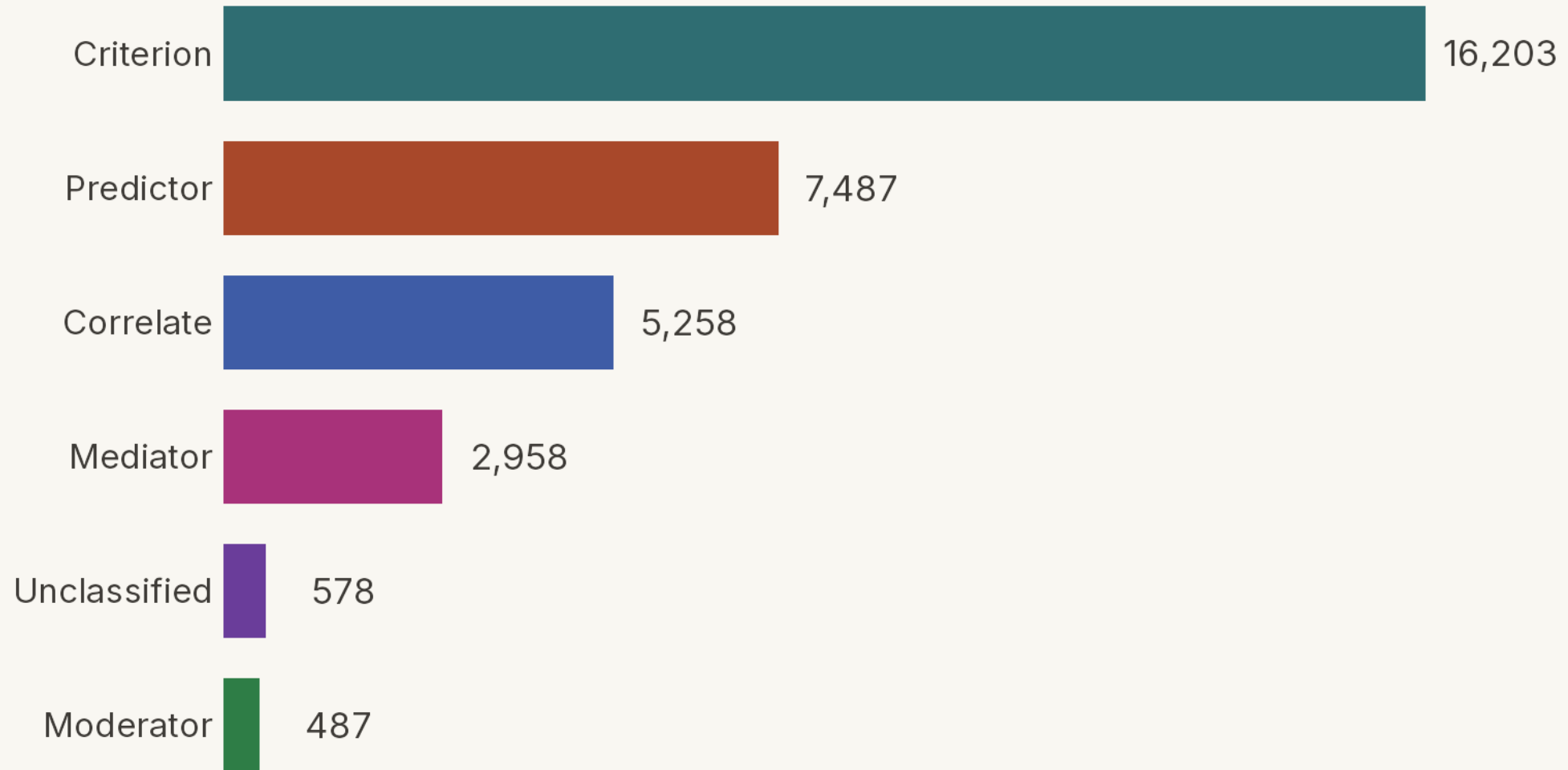
How 'organizational commitment' is used

Annual count of abstracts using commitment in each role



How 'organizational commitment' is used — all six roles

Total studies coded in each role — a study can hold several, so bars exceed the 22,484 coded



How many distinct variables did we extract?

After lightly combining spelling and plural variants — before construct grouping

Predictors of commitment

14,390

distinct variables

Outcomes of commitment

3,320

distinct variables

Light combining is cosmetic — it merges only spellings and plurals; the real reduction comes from construct grouping.

One construct, every language

For every study, the AI extracts the variables it examines — reading the abstract in its own language — and records each under one English label, so studies in any language pool into the same construct.

■ As written across languages

job satisfaction

工作满意度

satisfação no trabalho

iş tatmini

satisfacción laboral

satisfaction au travail

직무 만족

الرضا الوظيفي

English

Chinese

Portuguese

Turkish

Spanish

French

Korean

Arabic

■ The unified label

job_satisfaction

one English label · **2,085** studies pooled

The alternative — human coding — would either **miss the non-English articles** or require **hiring multiple language-proficient coders**. The AI needs neither.

How did we combine the variable labels?

Light combining — merging only plural/spelling variants, before construct grouping

BOX

1

Extract the raw labels

Direct extraction of the coded text,
“as is.”

BOX

2

Merge plural / spelling variants

Combine simple plural and spelling variants (e.g., “policies” → “policy”).

BOX

3

Group by meaning

Group labels with the same meaning into broad families (shown next).

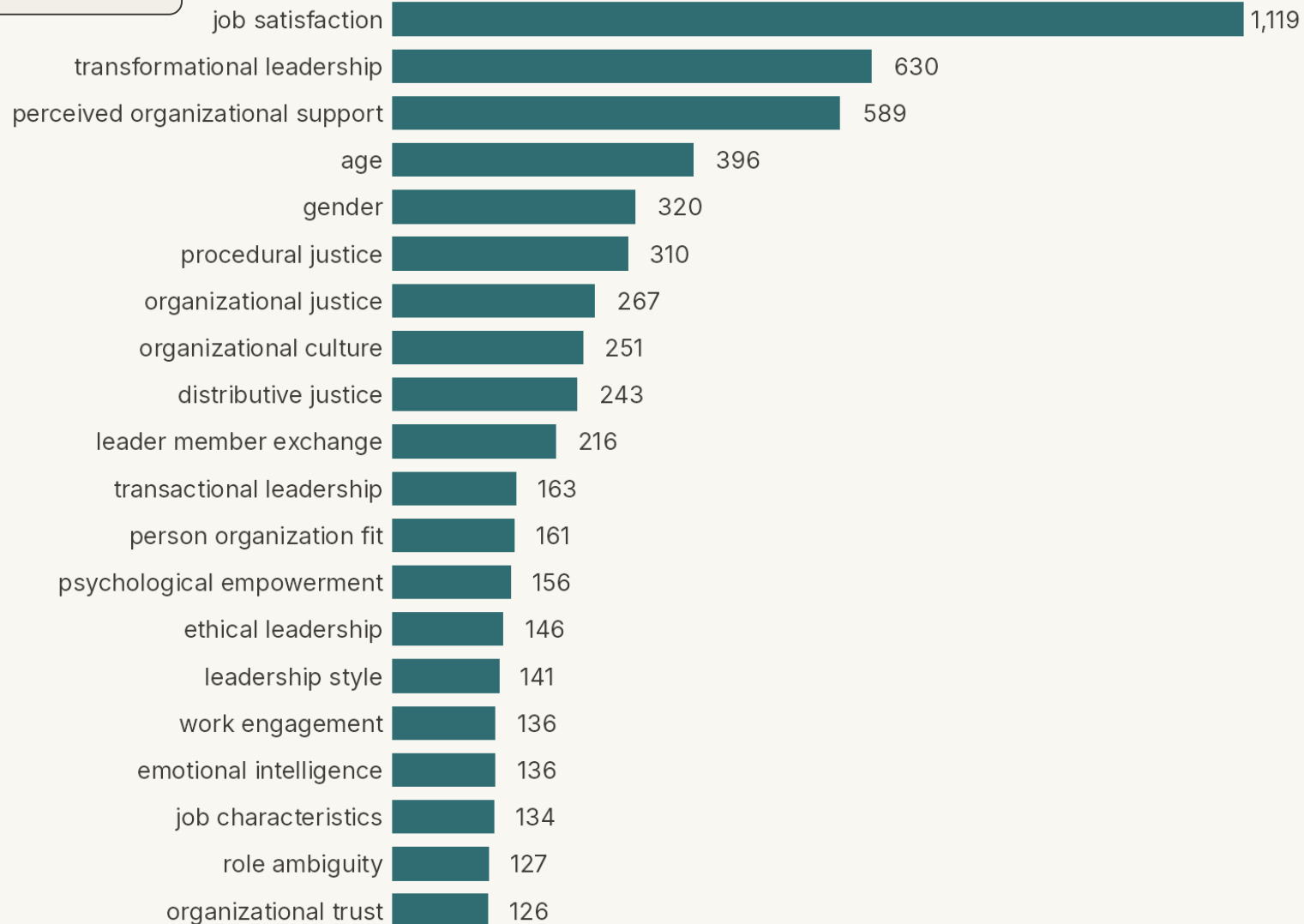
Conservative by design — spelling merges are verified against the coded labels and logged, and the meaning-based grouping (Box 3) is a first pass, still under coauthor review.

Box 2

Merge plural / spelling variants

Top 20 predictors of organizational commitment

Of 14,390 distinct predictor variables (light combining) — bar length = distinct studies naming each



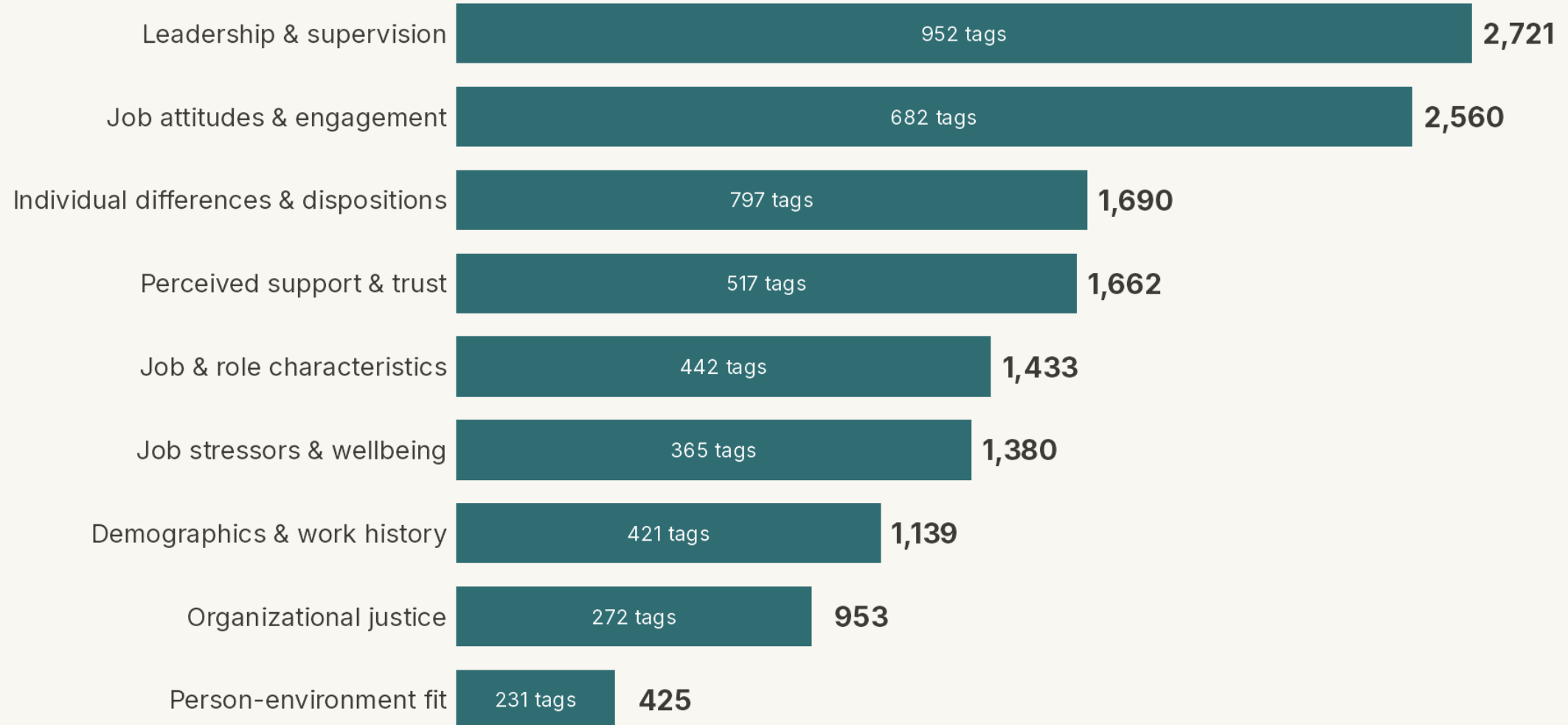
Box 3

Group by meaning

Predictors of organizational commitment, in nine categories

Bar & end label = distinct studies naming ≥ 1 predictor in the category.

Inside each bar = number of variables (tags) collapsed into it. ("Other" not shown.)

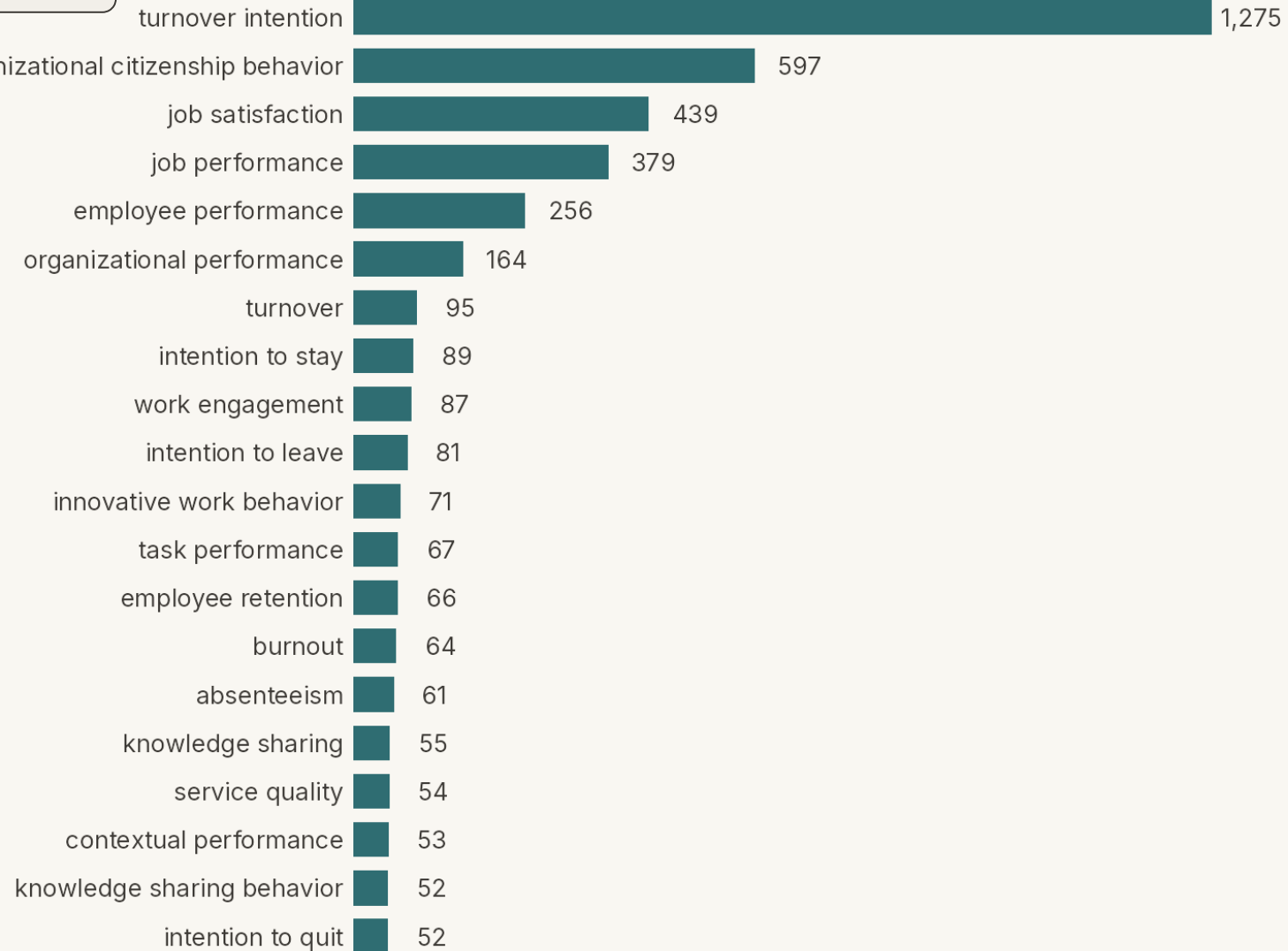


Box 2

Merge plural / spelling variants

Top 20 outcomes of organizational commitment

Of 3,320 distinct outcome variables (light combining) — bar length = distinct studies naming each



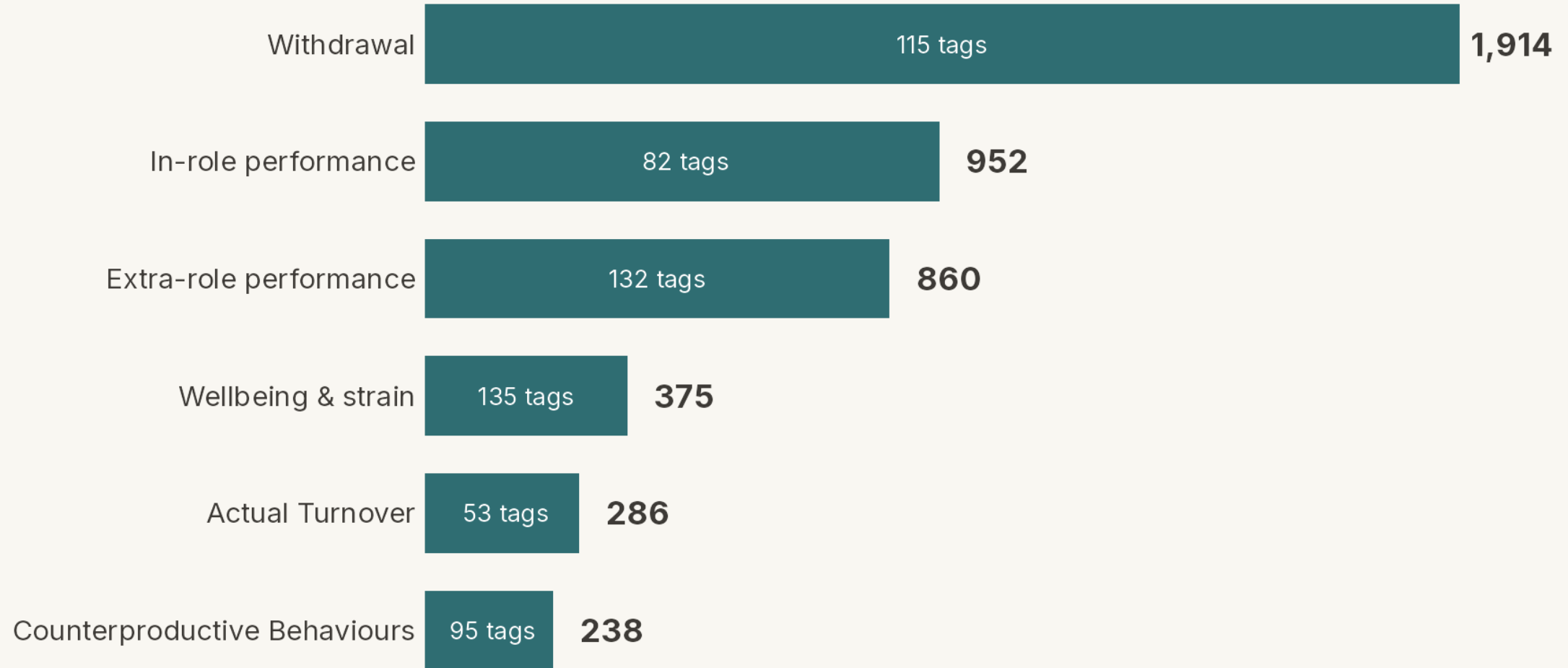
Box 3

Group by meaning

Outcomes of organizational commitment, in six categories

Bar & end label = distinct studies naming ≥ 1 outcome in the category.

Inside each bar = number of variables (tags) collapsed into it.



Top 20 correlates of organizational commitment

Of 4,030 distinct correlate variables (light combining) — bar length = distinct studies naming each



FROM ABSTRACTS TO ANALYSIS

Next Steps

The work ahead, in four steps.

How accurate is AI coding? — from prior research

The labour-intensive part is the coding itself — reading each study and pulling out every sample size, mean, correlation, and reliability by hand. In earlier published validation, the AI was compared against the expert human teams who did exactly that.

■ Data extraction — the coding itself

97.0%

AI accuracy — vs **92.1%** for expert human coders

4 meta-analyses · 632 studies · 20,476 data points

■ What the AI is being compared to

One of the four meta-analyses, single-coded by hand:

130 hrs → 4 hrs
one coder ≈ 32× faster the AI

12,025 numbers from 390 studies — one coder, about 20 min a study.

[Mehr, A., Howard, J. L., Nouroozi, C., Khorramnazari, B., Banks, G. C., Tay, L., Cuijpers, P., Miguel, C., Harrer, M., Meyer, J. P., Stanley, D. J., Wang, X., Luo, G., Huo, B., Liu, J., & Rousseau, D. M. \(2026\). *Improving Evidence Synthesis with Artificial Intelligence* \[Preprint\].](#)

Next steps

From the coded abstracts to a completed meta-analysis — four steps from here.

1

Download the articles

Retrieve the full texts behind the coded abstracts.

2

Code the articles

Extract effect sizes and study details from each.

3

Formalize the variable taxonomies

Finalize the construct groupings from these results.

4

Begin analyses

Run the meta-analytic models.

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Toward an AI-assisted Comprehensive Meta-analysis of the Organizational Commitment Literature

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Foundational works

1 OF 2

The cited-reference search retrieved every record citing any of these 14 works

Allen, N. J., & Meyer, J. P. (1990). The measurement and antecedents of affective, continuance and normative commitment to the organization. *Journal of Occupational Psychology*, 63(1), 1–18.

Allen, N. J., & Meyer, J. P. (1996). Affective, continuance, and normative commitment to the organization: An examination of construct validity. *Journal of Vocational Behavior*, 49(3), 252–276.

Cook, J., & Wall, T. (1980). New work attitude measures of trust, organizational commitment and personal need non-fulfilment. *Journal of Occupational Psychology*, 53(1), 39–52.

Klein, H. J., Cooper, J. T., Molloy, J. C., & Swanson, J. A. (2014). The assessment of commitment: Advantages of a unidimensional, target-free approach. *Journal of Applied Psychology*, 99(2), 222–238.

Meyer, J. P., & Allen, N. J. (1984). Testing the “side-bet theory” of organizational commitment: Some methodological considerations. *Journal of Applied Psychology*, 69(3), 372–378.

Meyer, J. P., & Allen, N. J. (1991). A three-component conceptualization of organizational commitment. *Human Resource Management Review*, 1(1), 61–89.

Meyer, J. P., & Allen, N. J. (1997). *Commitment in the workplace: Theory, research, and application*. Sage Publications.

Foundational works

2 OF 2

The cited-reference search retrieved every record citing any of these 14 works

Meyer, J. P., Allen, N. J., & Smith, C. A. (1993). Commitment to organizations and occupations: Extension and test of a three-component conceptualization. *Journal of Applied Psychology*, 78(4), 538–551.

Meyer, J. P., Stanley, D. J., Herscovitch, L., & Topolnytsky, L. (2002). Affective, continuance, and normative commitment to the organization: A meta-analysis of antecedents, correlates, and consequences. *Journal of Vocational Behavior*, 61(1), 20–52.

Mowday, R. T., Porter, L. W., & Steers, R. M. (1982). *Employee–organization linkages: The psychology of commitment, absenteeism, and turnover*. Academic Press.

Mowday, R. T., Steers, R. M., & Porter, L. W. (1979). The measurement of organizational commitment. *Journal of Vocational Behavior*, 14(2), 224–247.

O'Reilly, C. A., III, & Chatman, J. (1986). Organizational commitment and psychological attachment: The effects of compliance, identification, and internalization on prosocial behavior. *Journal of Applied Psychology*, 71(3), 492–499.

Porter, L. W., Steers, R. M., Mowday, R. T., & Boulian, P. V. (1974). Organizational commitment, job satisfaction, and turnover among psychiatric technicians. *Journal of Applied Psychology*, 59(5), 603–609.

Powell, D. M., & Meyer, J. P. (2004). Side-bet theory and the three-component model of organizational commitment. *Journal of Vocational Behavior*, 65(1), 157–177.